

The Empirics of Financial Markets

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The Term Structure of Interest Rates

The Term Structure of Interest Rates

- **Basic definitions and summary statistics**

- Yields, holding returns, risk premia, and forward rates

- **The expectation hypothesis (EH)**

- Theory, implications and testing

- **Modeling the yield curve**

- Factor models (purely statistical approach, may allow for arbitrage trades)
- No-arbitrage models (EH, affine TSMs)

The Term Structure of Interest Rates

Basic definitions

Zero coupon bond (ZCB, discount bond): claim that does not promise any payments (coupons) during its time to maturity, but promises to pay 1 unit of account (or equivalently 100% of the face value) at its time of maturity.

ZCBs are basic building blocks in finance!

Zero coupon bond price: P_{nt} : the price at time t of a ZCB that matures n periods in the future (typically months or years) and yields at that time a face value of 1 unit of account.

The Term Structure of Interest Rates

Zero coupon bond yield: Y_{nt} or y_{nt} :

- *Discretely compounded* yield to maturity, Y_{nt} :

$$P_{nt} (1 + Y_{nt})^n = 1 \quad \Leftrightarrow \quad P_{nt} = \frac{1}{(1 + Y_{nt})^n}.$$

i.e. the steady rate at which the price P_{nt} should increase if the ZCB is to be worth 1 unit of account at time $t + n$. Nonlinear and inverse relation between prices and yields (one-to-one mapping)!

- *Continuously compounded* or *log* yield to maturity y_{nt} :
 $y_{nt} \equiv \ln(1 + Y_{nt})$. Linear relation between log yield y_{nt} and log price p_{nt} , $p_{nt} \equiv \ln(P_{nt})$:

$$p_{nt} = -ny_{nt} \quad \Leftrightarrow \quad y_{nt} = -\frac{1}{n}p_{nt}.$$

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Term structure of interest rates or ZCB yield curve: plots the yields of bonds against their remaining time to maturity at a specific point in time. The underlying bonds have to have the same credit risk, liquidity risk, and tax treatment.

Probably one of the most important concepts and curves in the finance literature:

- **Central banks:** care about how ST interest rates are "transmitted" into LT interest rates
- **Public debt** (re)financing
- **Macroeconomic analyst:** predict inflation, recessions, etc.
- **Risk managers:** hedge their exposures
- **Pricing** of derivatives, interest rate swaps, etc.

The Term Structure of Interest Rates

Basic definitions

(Holding period) returns: $R_{n,t+1}$ return from buying n -period zero coupon bond at time t and selling it at time $t + 1$:

$$1 + R_{n,t+1} = \frac{P_{n-1,t+1}}{P_{nt}}$$

log (holding) return $r_{n,t+1} \equiv \ln(1 + R_{n,t+1})$:

$$\begin{aligned} r_{n,t+1} &= \ln(P_{n-1,t+1}/P_{nt}) = p_{n-1,t+1} - p_{nt} \\ &= y_{nt} - (n-1)(y_{n-1,t+1} - y_{nt}), \end{aligned}$$

- Unlike the log *yield to maturity*, the log *return* is *not* known in advance because it depends on the uncertain price at which the ZCB can be sold in the next period.
- Returns will equal yields *only if* long yields were to remain unchanged. If long yields rise, **capital loss** on the ZCB and the holding return is less than the yield (leverage role of the time to maturity). Similarly, if the long yield falls, **capital gain** and the return exceeds the yield.

Given that

$$p_{nt} = -r_{n,t+1} + p_{n-1,t+1}$$

we find, by repeated substitution, that:

$$p_{nt} = - \sum_{i=0}^{n-1} r_{n-i,t+1+i}$$

hence

$$\begin{aligned} y_{nt} &= -\frac{1}{n} p_{nt} \\ &= \frac{1}{n} \sum_{i=0}^{n-1} r_{n-i,t+1+i} \end{aligned}$$

The ZCB yield on a long bond is equal to the average return per period, if the bond is held to maturity! (Such a statement does not hold for coupon bonds, unless one is willing to make the extra assumption that the intermediate coupons can be invested at the initial yield to maturity.)

The Term Structure of Interest Rates

Basic definitions

Excess returns: ZCB returns are predictable over the life-time of the bond, since we know with certainty ("risk free") that the price will be "pulled to par" at maturity. Any unexpected price change that may occur in the near future must be offset by an opposite price change in the more distant future to bring the price back to face value on the maturity date. Returns are negatively serially correlated and hence forecastable, especially so when the time of maturity is approaching ($r_{1,t+1} = y_{1t}$). Hence, focus on excess returns:

$$er_{n,t+1} \equiv r_{n,t+1} - r_{1,t+1} = (y_{nt} - y_{1t}) - (n-1)(y_{n-1,t+1} - y_{nt})$$

- Excess returns equal yield spreads corrected for future changes in long yields.
- Leverage effect of changes in long yields. Yield spreads can be biased predictors of excess returns. A 7% 30 year ZCB only has to rise with 10.5 basis points over the next year to earn exactly the same 1 year return as a 4% 1 year ZCB.

The Term Structure of Interest Rates

Basic definitions

Expected one-period log holding return:

$$\begin{aligned} E_t \{r_{n,t+1}\} &= y_{nt} - (n-1) E_t \{y_{n-1,t+1} - y_{nt}\} \\ &= ny_{nt} - (n-1) E_t \{y_{n-1,t+1}\} \end{aligned}$$

Define the **(log) risk premium** as the expected excess return:

$$\phi_{n,t} \equiv E_t \{r_{n,t+1}\} - y_{1t}.$$

$$\begin{aligned} \phi_{n,t} &= \{y_{nt} - y_{1t}\} - (n-1) E_t \{y_{n-1,t+1} - y_{nt}\} \\ &= ny_{nt} - (n-1) E_t \{y_{n-1,t+1}\} - y_{1t} \end{aligned}$$

The Term Structure of Interest Rates

Basic definitions

Forward rates (implied by ZCB yields):

Lock in or guarantee future interest rates today, by buying and selling ZCBs. The guaranteed future interest rates are referred to as forward rates.

For example, lock in the interest rate between $t + n$ and $t + n + 1$:

- Buy a single $n + 1$ -maturity ZCB at time t that will give you a payoff of 1 at time $t + n + 1$.

Postpone the financing of this investment until period $t + n$, by selling a specific number of n ZCBs at time t . The cash flows at time t , $t + n$, and $t + n + 1$ are as follows:

	t	$t + n$	$t + n + 1$
Buy a bond with maturity $n + 1$	$-P_{n+1,t}$		1
Finance with sale of n -maturity bonds	$\left\{ \frac{P_{n+1,t}}{P_{nt}} \right\} P_{nt}$	$-\frac{P_{n+1,t}}{P_{nt}}$	
Net	0	$-\frac{P_{n+1,t}}{P_{nt}}$	1

The return on the above transaction is defined as the forward rate:

$$(1 + F_{nt}) = \frac{1}{\left(\frac{P_{n+1,t}}{P_{nt}}\right)} = \frac{P_{nt}}{P_{n+1,t}}$$

or in logs:

$$\begin{aligned} f_{nt} &= \ln(1 + F_{nt}) \\ &= p_{nt} - p_{n+1,t} \\ &= -ny_{nt} + (n+1)y_{n+1,t} \\ &= y_{nt} + (n+1)(y_{n+1,t} - y_{nt}) \end{aligned}$$

- Forward rate is above (below) the corresponding yield when the yield curve is upward (downward) sloping.
- The forward rate is equal to the yield when the yield curve is flat.

Coupon bond yields

Coupon bonds can be considered to be portfolios of ZCBs with payoffs and maturities that match the coupon payments:

$$\begin{aligned}P_{nt}^c &= \frac{c}{1 + Y_{1t}} + \frac{c}{(1 + Y_{2t})^2} + \dots + \frac{1 + c}{(1 + Y_{nt})^n} = \\&= \sum_{i=1}^n \frac{c}{(1 + Y_{it})^i} + \frac{1}{(1 + Y_{nt})^n}.\end{aligned}$$

The yield to maturity of this coupon bond, $\tilde{Y}_{nt}$, can be derived from:

$$\begin{aligned}P_{nt}^c &= \frac{c}{1 + \tilde{Y}_{nt}} + \frac{c}{(1 + \tilde{Y}_{nt})^2} + \dots + \frac{1 + c}{(1 + \tilde{Y}_{nt})^n} = \\&= \sum_{i=1}^n \frac{c}{(1 + \tilde{Y}_{nt})^i} + \frac{1}{(1 + \tilde{Y}_{nt})^n}.\end{aligned}$$

It is the interest rate that equates the sum of the net present values of its cash flows to its price. Relatively easy when cash flows are certain.

Complex when cash flows are uncertain (for example: prepayments of loans).

The Term Structure of Interest Rates

Descriptive statistics

- McCulloch and Kwon (1993) US government yield data set.
- Continuously compounded yields to maturity on ZCBs, observed at monthly intervals, and expressed as annual percentage points.
- Sample 1952:01-1991:02, totaling 470 monthly data points.
- Cross-section of 21 yields, with maturities (n) 1, 2, ..., 12, 24, 36, 48, 60, 72, 84, 96, 108, and 120 months.

Descriptive statistics: yields and yield spreads (US data: 1952:01-1991:02)

Maturities are measured in months. For each time series, the table reports the sample mean (Mean) and standard deviation (Std), the minimum (Min) and maximum (Max), the skewness (Skew) and kurtosis (Kurt), the Jarque-Bera (1980) test statistic for normality (JB), and the monthly autocorrelation (Auto), using the 470 monthly observations. The units of the entries are specified below the column headings. The underlying data are ZCB yields. The superscript stars (^{*}) in columns Skew, Kurt and JB denote rejection of the null hypotheses (no skewness, no kurtosis, and normality) at the 95% confidence level. The null hypotheses are, respectively, H_0 : no skewness (two-sided), H_0 : no excess kurtosis (two-sided), and H_0 : normality. Note that the JB test is prone to type II error, when the underlying process is actually normally distributed.

n (mon.)	Mean (% pa)	Std (% pa)	Min (% pa)	Max (% pa)	Skew	Kurt (-3)	JB	Auto (mon)
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Panel A: Yields: y_{nt}

1	5.314	3.064	0.249	16.210	0.88*	0.8*	72*	0.976
3	5.640	3.143	0.615	15.999	0.85*	0.7*	66*	0.982
12	6.079	3.168	0.847	16.345	0.73*	0.3	43*	0.983
24	6.272	3.124	1.149	16.145	0.66*	0.1	34*	0.986
36	6.386	3.087	1.412	15.825	0.62*	-0.1	30*	0.988
48	6.467	3.069	1.595	15.847	0.61*	-0.1	30*	0.989
60	6.531	3.056	1.770	15.696	0.60*	-0.2	29*	0.990
120	6.683	3.013	2.341	15.065	0.53*	-0.5*	27*	0.993

Panel B: yield spreads: $y_{nt} - y_{1t}$

3	0.326	0.303	-0.537	2.011	2.03*	7.0*	1274*	0.353
12	0.765	0.593	-1.524	3.655	1.27*	4.9*	592*	0.686
24	0.959	0.796	-2.057	4.378	0.53*	2.6*	150*	0.793
36	1.073	0.927	-2.389	4.744	0.27*	1.9*	80*	0.832
48	1.154	1.011	-2.672	4.829	0.10	1.5*	46*	0.851
60	1.217	1.078	-2.896	4.960	0.03	1.3*	33*	0.864
120	1.369	1.237	-3.323	5.269	-0.09	0.8*	13*	0.887

n	Mean (% pa)	Std (% pa)	Min (% pa)	Max (% pa)	Skew	Kurt (-3)	JB	Auto (mon)
Panel C: Monthly changes in yields: $y_{n,t+1} - y_{n,t}$								
3	-0.119	0.586	-4.717	2.166	-2.1*	14*	4339*	0.136
12	-0.014	0.555	-4.274	3.176	-0.7*	12*	2746*	0.169
24	0.011	0.488	-3.646	3.143	-0.4*	11*	2501*	0.132
36	0.011	0.442	-2.863	2.747	-0.0	8*	1247*	0.100
48	0.011	0.410	-2.310	2.449	0.0	6*	762*	0.088
60	0.012	0.383	-1.943	2.183	0.1	5*	497*	0.077
120	0.012	0.310	-1.309	1.380	-0.2	3*	205*	0.069
Panel D: Monthly excess returns: $r_{n,t+1} - r_{1,t+1}$								
3	0.564	1.222	-4.543	9.604	2.1*	13*	3747*	0.179
12	0.918	6.217	-35.078	47.224	0.7*	11*	2442*	0.183
24	0.709	11.326	-72.845	82.933	0.3*	11*	2262*	0.141
36	0.682	15.591	-97.113	98.380	-0.0	8*	1153*	0.109
48	0.614	19.402	-116.229	106.206	-0.1	6*	732*	0.097
60	0.526	22.733	-129.944	111.949	-0.1	5*	485*	0.086
120	-0.048	37.063	-165.188	152.718	0.2	3*	204*	0.077

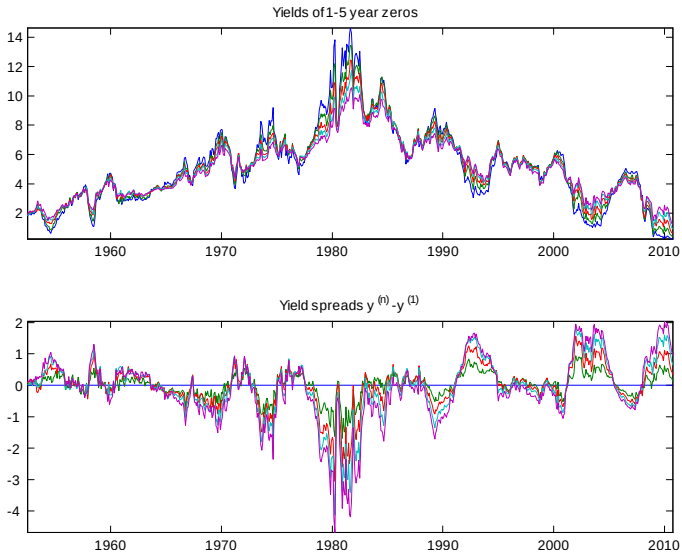


Figure: US ZCB nominal yield and yield spread dynamics (1952 -2010)

Observations

- Average yield curve: upward sloping, concave (inverted yield curve 1973 and early 80s)
- Yields persistent in levels (unit root?)
- Volatility curve: hump shaped, long yields not constant
- Nonnormalities: excess kurtosis
- Yield changes
- Excess return: low mean, high standard deviation, decomposition: interpretation, standard errors
- Correlation matrix yields/yield changes: factor model

The Term Structure of Interest Rates

The (pure) expectations hypothesis

Link long yields to short yields: rewrite the risk premium equation:

$$y_{nt} = \frac{1}{n}y_{1t} + \frac{n-1}{n}E_t \{y_{n-1,t+1}\} + \frac{1}{n}\phi_{n,t}.$$

Substitution in itself gives:

$$y_{nt} = \frac{1}{n}y_{1t} + \frac{1}{n}\phi_{n,t} + \frac{n-1}{n} \left\{ \frac{1}{n-1}E_t \{y_{1,t+1}\} + \frac{n-2}{n-1}E_t \{y_{n-2,t+2}\} + \frac{1}{n-1}E_t \{\phi_{n-1,t+1}\} \right\},$$

Iterate:

$$y_{nt} = \frac{1}{n} \sum_{i=0}^{n-1} E_t \{y_{1,t+i}\} + \Phi_{n,t}$$

where $\Phi_{n,t}$ is defined as:

$$\Phi_{n,t} = \sum_{i=0}^{n-1} \frac{1}{n} E_t \{\phi_{n-i,t+i}\}.$$

- **Fundamental relationship!** y_{nt} is a sum of expected short term interest rates and a sum of log risk premia. The equation does not involve any assumptions and holds by virtue of the definition of $\phi_{n,t}$. The $\Phi_{n,t}$ -term is referred to as the risk or term premium, reflecting the extra yield risk averse individuals require for holding the riskier long bonds rather than short bonds.
- As soon as a specification for future expected short rates and for the term premia is added, a direct relationship between the current long rates and observable variables can be obtained.
- **Expectations hypothesis** of the term structure of interest rates: term premium $\Phi_{n,t}$ is time-invariant, i.e. $\Phi_{n,t} = \Phi_n$.
- **Pure expectations hypothesis:** all risk premia are assumed to be zero ($\phi_{n,t} = 0$, for all n, t). It states that the long interest rate equals the average of the relevant expected future short rates. When yield spreads are positive, it is expected that short rates will rise over the life of the long bond, and vice versa.

The Term Structure of Interest Rates

Tests of the EH with OLS

We will test term structure models where the term premium $\Phi_{n,t}$ is assumed to be time-invariant. We do this by constructing two regression-based tests which are often reported in the literature.

Test EH empirically: test whether

$$y_{nt} = \frac{1}{n} \sum_{i=0}^{n-1} E_t \{y_{1,t+i}\} + \Phi_n,$$

holds for some constants Φ_n . Problem: expectations on the right hand side are generally unobserved. Derive two testable relationships.

The Term Structure of Interest Rates

Tests of the EH with OLS

Test 1: Test of long yield changes over the life of the short bond

In case the log risk premium ϕ_n is time-invariant, the equation becomes:

$$y_{nt} = \frac{1}{n}y_{1t} + \frac{n-1}{n}E_t\{y_{n-1,t+1}\} + \frac{1}{n}\phi_n$$

Construct a regression equation: rewrite:

$$E_t\{y_{n-1,t+1}\} - y_{nt} = -\frac{1}{n-1}\phi_n + \frac{1}{n-1}(y_{nt} - y_{1t})$$

If this equation holds, then the equation

$$y_{n-1,t+1} - y_{nt} = \alpha_0 + \alpha_1 \frac{1}{n-1}(y_{nt} - y_{1t}) + \varepsilon_{t+1}$$

should hold with $\alpha_0 = -\frac{1}{n-1}\phi_n$, and $\alpha_1 = 1$. In sum, we run an OLS regression of $y_{n-1,t+1} - y_{nt}$ on a constant and the adjusted spread $\frac{1}{n-1}(y_{nt} - y_{1t})$ and test whether the slope coefficient α_1 is equal to 1.

The Term Structure of Interest Rates

Tests of the EH with OLS

Test 2: Test of short yield changes over the life of the long bond

A second way to test the EH using OLS: subtract the short yield from left and right hand sides:

$$y_{nt} - y_{1t} = \frac{1}{n} \sum_{i=0}^{n-1} E_t \{y_{1,t+i}\} - y_{1t} + \Phi_n.$$

If this equation holds, then

$$\frac{1}{n} \sum_{i=0}^{n-1} y_{1,t+i} - y_{1t} = \beta_0 + \beta_1 (y_{nt} - y_{1t}) + \varepsilon_{t+n-1}$$

should also hold with $\beta_0 = -\Phi_n$ and $\beta_1 = 1$. By construction we have an overlapping samples problem. Use standard errors which correct for the presence of autocorrelation. In our case we will use Newey-West standard errors, to correct for autocorrelation and heteroskedasticity. Run OLS and check whether the slope coefficient β_1 is equal to 1.

The Term Structure of Interest Rates

Tests of the EH with OLS

Test results on US data

To illustrate the two regression based tests explained above, we run the above OLS regressions using data of different yields to maturity for the United States.

- McCulloch and Kwon (1993) data set.
- There is a small problem in estimating the first equation for n equal to 36, 60 and 120: we have to calculate $y_{n-1,t+1} - y_{nt}$ and use it as a dependent variable. We approximate it by using $y_{n,t+1} - y_{nt}$ instead of $y_{n-1,t+1} - y_{nt}$, as is usually done in papers on this topic.
- Regression-based tests of the expectations hypothesis

s.e. denotes traditional standard errors, s.e.^a denotes White standard errors in case of the regression (1) and Newey-West standard errors with $n-1$ lags in case of the (2) regression. ^b signals that the approximation $y_{n,t+1} \approx y_{n-1,t+1}$ is used.

n (months)		Equation (1)		Equation (2)	
		α_0	α_1	β_0	β_1
3	coeff.	−0.0008	−0.0183	0.0012	0.4488
	s.e.	0.0003	0.1624	0.0003	0.0645
	s.e. ^a	0.0004	0.2548	0.0003	0.1293
6	coeff.	0.0003	−0.7781	−0.0013	0.3119
	s.e.	0.0003	0.2668	0.0005	0.0721
	s.e. ^a	0.0004	0.3862	0.0006	0.1387
12	coeff.	0.0007	−1.2908	−0.0011	0.2672
	s.e.	0.0003	0.4182	0.0006	0.0723
	s.e. ^a	0.0004	0.5266	0.0009	0.1649
36	coeff.	0.0006 ^b	−1.7257 ^b	−0.0016	0.4083
	s.e.	0.0003	0.6965	0.0010	0.0765
	s.e. ^a	0.0004	1.1516	0.0030	0.2875
60	coeff.	0.0006 ^b	−2.4332 ^b	−0.0024	0.5319
	s.e.	0.0002	0.8840	0.0011	0.0793
	s.e. ^a	0.0003	1.4548	0.0029	0.4023
120	coeff.	0.0005 ^b	−4.0408 ^b	−0.0040	1.3771
	s.e.	0.0002	1.2706	0.0006	0.0465

The Term Structure of Interest Rates

Tests of the EH with OLS

Regression-based test of the expectations hypothesis (Campbell and Shiller, 1991)

n (months)		Regression 1 ^a	Regression 2 ^b
		Slope Coefficient ^c	Slope Coefficient ^c
3	coeff.	-0.176	0.446
	s.e.	0.362	0.190
6	coeff.	-1.029	0.237
	s.e.	0.537	0.167
12	coeff.	-1.381	0.161
	s.e.	0.683	0.228
36	coeff.	-2.239 ^d	0.614
	s.e.	1.444	0.230
60	coeff.	-3.099 ^d	1.232
	s.e.	1.749	0.182
120	coeff.	-5.024 ^d	1.157
	s.e.	2.316	0.094

The Term Structure of Interest Rates

Tests of the EH with OLS

Estimates are not in line with what we expect from the theory!

- **First regression:** slope coefficients negative and significantly different from one. Constant terms all positive and many significantly different from zero.
- **Second regression:** somewhat more in line with the theory. At least positive slope estimates, but at short end rejections, at long end, no statistically significant rejection (for testing, use the NW s.e.). The estimates for the constant are in all but the first case negative as could be expected, and in most cases significantly different from zero.
- Campbell and Shiller test: similar results. In the first regression each estimated slope coefficient is negative and significantly different from one. The results of the second regression are somewhat better.
- Apparent paradox: slope of the term structure almost always gives a forecast in the wrong direction for the short-term change in the yield on the longer bond, but gives a forecast in the right direction for the long-term changes in the short rates.

The Term Structure of Interest Rates

Tests of the EH with OLS

Explanations for EH rejection (Bekaert, Hodrick and Marshall, 1997):

- Extreme **biases in the small-sample distributions** of the standard tests due to the extreme **persistence in short interest rates**. However, correcting for the small sample bias in the test, the conclusions remain the same.
- **Time-varying risk premia** and the use of panel data. Dai and Singleton (2002) and Maes (2003) use panel data estimation methods and multi-factor term structure models: convincing evidence that incorporating model-implied time-varying risk premia reveals the predictive content of the yield spread with respect to short horizon changes in long rates.
- **"Peso-problems"**; it is always possible that the ex post realizations of the yields available in the data set are not reliable proxies for the expectations.
- **Failure of rational expectations** in sample

The Term Structure of Interest Rates

Predicting recessions with the yield curve

- Risk premia are typically counter-cyclical: high in recessions and low in booms. Hence, long yields are typically high in recessions and low in boom periods.
- Central banks typically "lean against the wind" or "take away the punch bowl when the party gets going", i.e. they raise short term interest rates in booms and lower them in recessions. A monetary tightening both slows down the economy and flattens the yield curve.
- Hence, recession end-points typically coincide with upward sloping yield curves and boom end-points typically see flat or even inverted yield curves.
- Evidence is supportive of the predictive value of the yield slope on economic growth. Every US recession since 1965 was predicted by a negative slope within 6 quarters of the impending recession (with only one single false negative).
- Bernanke (2004) quote in lecture notes

The Term Structure of Interest Rates

Modelling the yield curve

- **Factor models** statistically *summarize* the information at any point in time.
 - A small number of sources of systematic risk appear to underlie the pricing of the myriad of tradable financial assets.
 - Yield curve models employ a structure that consists of a small set of *factors* and the associated *factor loadings* that relate yields of different maturities to these factors.
 - A first approach imposes structure on the estimated factors only. The factors could be the first few principal components, while the loadings are relatively unrestricted.
- Alternative approach: **no-arbitrage dynamic latent factor model** (finance research). The most common subclass of these models postulates the absence of arbitrage opportunities and the existence of a linear or affine relation between the factors and the bond yields.

The Term Structure of Interest Rates

Factor models

Apply an **eigenvalue-eigenvector** or **spectral decomposition** to any symmetric matrix Σ :

$$\Sigma = V\Lambda V'$$

with V the matrix with orthonormal eigenvectors stacked as columns ($V'V = I$), and Λ the diagonal matrix of ordered eigenvalues (λ_i) of Σ on the diagonal.

Study a large number of yields by a small number of orthogonal factors:

$$\text{Cov}(y, y') = \Sigma = V\Lambda V'$$

may be interpreted as reflecting the fact that yields are generated from a number of uncorrelated variables x via a loading matrix V :

$$y = Vx$$

where it follows that

$$\text{Cov}(x, x') = \Lambda = \textit{diagonal}$$

The Term Structure of Interest Rates

Factor models

The yields are generated by a factor model with uncorrelated factors x and regression coefficients on the factors given by the columns of V . The x are referred to as factors or principal components, while the (orthonormal) column vectors of V are referred to as the factor loadings, since they tell you how much y changes when a factor x moves.

It turns out that almost all variation in yields can be described in terms of a few factors, typically a "**level**", "**slope**", and "**curvature**" or "hump" factor. Knez, Litterman, and Scheinkman (1994) make the point with a formal maximum likelihood factor analysis, but the point can be illustrated with our more simple spectral decomposition of log yields.

Results from the spectral decomposition of the yield data set that consists of the 1, 2, 3, 4, and 5 year zero coupon bond yields covering the period 1952:05-2004:12 (631 monthly observations).

The first factor accounts for most of the movement in y .

Rather than to express this with the variance of the factors (Λ), we will express this through the loadings L of *unit variance factors* z ,
 $\text{Cov}(z, z') = I$, $z = \Lambda^{-1/2}x$:

$$\begin{aligned} y &= Lz \\ \Sigma &= LL' = \left(V\Lambda^{1/2}\right) I \left(V\Lambda^{1/2}\right)' \end{aligned}$$

- The loading vectors V and L reveal that the first factor can be interpreted as a **"level" factor**.
- The second factor is already less important in explaining the variability of yields. Its loading pattern implies that it can be labelled as **"slope" factor**.
- Similarly, the third factor can be labelled **"curvature" factor**.

	Principal components				
	1	2	3	4	5
$\sqrt{\Lambda_{i,i}}$	5.8581	0.5794	0.1000	0.0744	0.0574
Maturity	V				
1	0.4623	-0.7346	-0.4765	0.1287	0.0543
2	0.4577	-0.2126	0.5889	-0.5377	-0.3306
3	0.4460	0.1215	0.4253	0.3799	0.6791
4	0.4387	0.3656	-0.0626	0.5390	-0.6159
5	0.4306	0.5164	-0.4912	-0.5093	0.2174
Labels	"level"	"slope"	"curvature"	"2/5"	"zigzag"
Maturity	L				
1	2.7081	-0.4256	-0.0476	0.0096	0.0031
2	2.6812	-0.1232	0.0589	-0.0400	-0.0190
3	2.6127	0.0704	0.0425	0.0283	0.0389
4	2.5702	0.2118	-0.0063	0.0401	-0.0353
5	2.5223	0.2992	-0.0491	-0.0379	0.0125

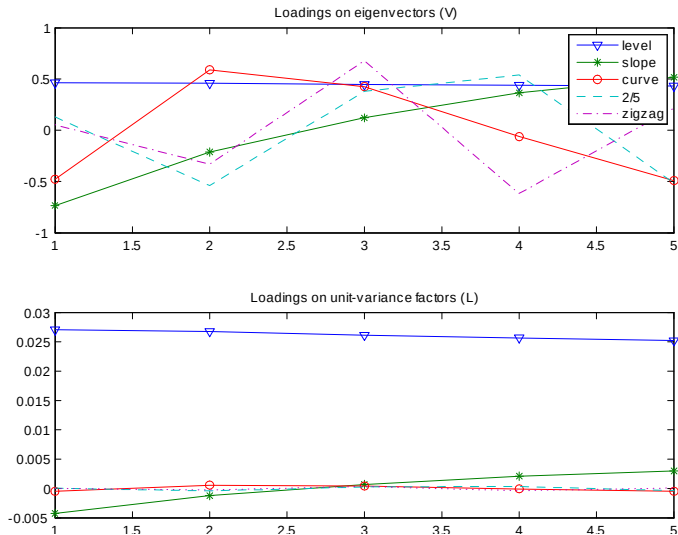


Figure: Implied loadings from a principal component analysis

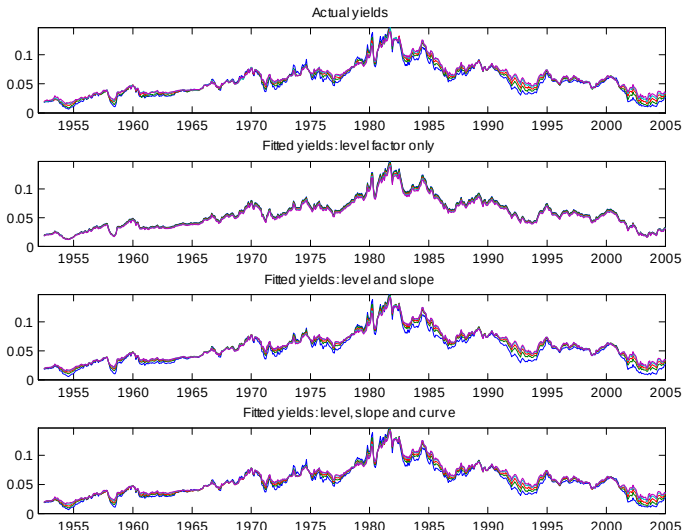


Figure: Yield data and model-implied yields where the 1, 2 and 3 most important principal components are used.

The Term Structure of Interest Rates

Linking the term structure to macroeconomic dynamics

Macro-finance models

- Macroeconomic perspective: the short-term interest rate is a policy instrument under the direct control of the central bank.
- Finance perspective: long rates are risk-adjusted averages of expected future short rates.

Thus, a joint macro-finance model may provide the most comprehensive understanding of the link between the term structure of interest rates and the dynamics of macroeconomic aggregates.

How can macroeconomic variables be linked to ZCB price dynamics?
(based on Ang and Piazzesi (2001)).

The Term Structure of Interest Rates

Conclusions

- Stylized facts of yields and returns
- Expectations hypothesis is rejected in many studies to date. Given the intuitive appeal and the bedrock status of the expectations theory, this is a puzzle that has been standing for decades as a challenge for term structure modelers.
- How the term structure can be modelled: purely statistical approaches (such as factor analysis) are popular, but do not provide a theoretical sound framework to filter the risk premia and they do not allow us to impose the no-arbitrage restrictions that characterize any reasonable general equilibrium in financial markets.
- Many researchers have found that the relationship between what is observed in bond markets and what is predicted by rational expectations is blurred by the presence of complex and time-varying risk premia, required by investors as compensation for being exposed to different sources of macroeconomic risks.